

Eliminating Casual Conversation in Audio Transcripts

CS4099 Project Final Report

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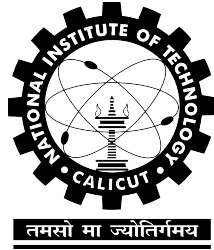


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CERTIFICATE

Certified that this is a bonafide record of the project work titled

ELIMINATING CASUAL CONVERSATION IN PODCASTS

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*of eighth semester B. Tech in partial fulfillment of the requirements for the
award of the degree of Bachelor of Technology in Computer Science and
Engineering of the National Institute of Technology Calicut*

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DECLARATION

We hereby declare that the project titled, **Eliminating Casual Conversation in Audio Transcripts**, is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or any other institute of higher learning, except where due acknowledgment and reference has been made in the text.

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Abstract

In the modern world, content is the primary source of information for a large part of the population. However, media sources often contain a significant amount of casual conversations that dilute core content, making it difficult for viewers and listeners to focus on relevant information. This report proposes an automated approach to refine audio transcripts by eliminating off-topic and informal discussions using Large Language Models (LLMs) and their zero-shot learning capabilities. By analyzing linguistic markers such as tone shifts, informal language, and topic divergence, our method efficiently distinguishes essential discussions from casual conversations. Additionally, we incorporate a fine-tuned model to enhance detection accuracy and classification, ensuring more structured, concise, and high-quality transcripts. This approach improves content clarity, coherence, and accessibility, enhancing the overall user experience.

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Chapter 1

Introduction

The initial aim of the project was to eliminate casual conversation in podcasts. However, on doing some research we came to the conclusion that there is only a minute amount of casual conversation in podcasts. Therefore, we generalized the problem and focused on other sources of media, as in the digital era today, content is a key source of information. We found a variety of content in various segments that have a lot of casual, off-topic conversations that dilute the core message, making it harder for audiences to focus on the relevant information. This report proposes an automated approach to refine audio transcripts by eliminating informal segments using Large Language Models (LLMs). By detecting tone shifts, informal language, and topic divergence, our method ensures concise and structured transcripts. A fine-tuned model further enhances accuracy, improving content clarity and accessibility. This report explores the methodology, effectiveness, and impact of our approach in streamlining audio transcripts for better content consumption.

Chapter 2

Literature Survey

The main objective of the literature survey was to find related papers based on text classification using LLMs and their zero-shot capabilities. In the field of text classification using LLMs, recent research has led to notable advancements. The following three studies are key contributions that have significantly influenced the development of hierarchical and adaptive text classification approaches, as well as the integration of progressive reasoning techniques in LLMs. These works provide a strong foundation for exploring the potential of LLMs in addressing complex classification challenges.

A. RGPT: Adaptive Boosting for LLM-based Classification: The RGPT framework, as described by Zhang et al. (2024), introduces an adaptive boosting approach that iteratively fine-tunes LLMs to improve text classification. RGPT focuses on creating specialized text classification models by ensembling base learners. By recurrently adjusting the distribution of training samples, the framework enhances the classification ability of LLMs beyond conventional prompt-based approaches. This work emphasizes pushing the boundaries of LLMs in classification, providing a competitive edge in handling zero-shot scenarios with minimal reliance on prompt engineering, marking a significant leap in adapting LLMs for diverse NLP tasks.

B. TELEClass: Taxonomy Enrichment and LLM-Enhanced Hierarchical Text Classification with Minimal Supervision Zhang et al. (2024) propose TELEClass, a system that enhances LLM performance for hierarchical text classification with minimal supervision. TELEClass automatically enriches label taxonomies with class-indicative terms, utilizing LLMs for both data annotation and generation. This framework addresses the challenge of applying LLMs in hierarchical settings, where large structured label spaces often hinder performance. By leveraging LLMs for core class identification and augmenting training data with pseudo-labeled documents, TELEClass achieves superior classification accuracy, especially in scenarios with limited labeled data. This method showcases the potential of LLMs in complex classification tasks that involve large taxonomies and fine-grained classes.

C. CARP: Clue And Reasoning Prompting Sun et al. (2023) introduced the CARP framework, which adapts LLMs to handle complex linguistic phenomena in text classification. CARP employs a progressive reasoning strategy in which LLMs are first prompted to collect clues (e.g., keywords, tones, semantic relations) and then reason through these clues to make a final classification decision. CARP’s ability to work in low-resource settings with few-shot learning significantly improves text classification performance, achieving state-of-the-art results on several benchmarks. By incorporating a fine-tuned search for k closest neighbors (kNN), CARP effectively bridges the gap between generalization capabilities of LLM and task-specific evidence.

While several frameworks explore innovative approaches to text classification, our project specifically draws inspiration from the CARP framework, which emphasizes clue-based reasoning for more accurate classification in low-context scenarios.

Chapter 3

Problem Definition

Audio transcripts from various media sources often contain casual or off-topic conversations that dilute the core message and reduce the clarity of the content. This makes it difficult for users to extract relevant information efficiently. Existing models struggle to accurately identify such segments, especially without a clear contextual understanding. This project aims to develop a method using Large Language Models (LLMs) to automatically detect and remove casual segments by leveraging reasoning-based classification and contextual cues, thereby enhancing the relevance and coherence of processed transcripts.

Chapter 4

Methodology

A. Design 1. Problem Understanding: The goal of this project is to filter out casual or off-topic segments from audio transcripts, which often reduce the relevance and clarity of downstream tasks like summarization and content extraction. These casual segments, such as small talk, filler phrases, or digressions, are common in YouTube videos and hinder the utility of transcripts. The challenge is that casual sentences are context-dependent and are not always obvious through syntax alone. In addition, there are no public datasets for this task, making it a low-resource classification problem. To address this, we draw inspiration from the CARP framework and use GPT-3.5-Turbo-Instruct to classify transcript segments. A Gemini-generated video summary is included with each prompt to provide context. The aim is to retain meaningful content by accurately identifying and filtering irrelevant segments.

2. Framework Selection and Initial Plan: After surveying various text classification methods using LLMs, we chose to base our approach on the CARP (Clue And Reasoning Prompting) framework. CARP is known for its multi-stage reasoning strategy, which first extracts clues and then performs classification, showing strong results in low-context, few-shot environments.

3. Initial Prompting Strategy: In the beginning, we adapted CARP’s three-stage prompting:

Clue Prompt: Ask the model to identify key linguistic clues from a transcript sentence.

Reasoning Prompt: Have the model reason using those clues.

Classification Prompt: Predict whether the sentence is casual or not.

However, this multi-stage approach was found to be inefficient and slow, especially when processing longer transcripts.

B. Implementation: 1. Model Selection: Due to the deprecation of text-davinci-003 used in the original CARP paper, we switched to GPT-3.5-Turbo-Instruct, which is optimized for instruction-following tasks and more cost-effective.

2. Final Prompting Strategy: We transitioned to a single-prompt approach to improve efficiency. Each prompt consists of:

A summary of the video, generated using Gemini, providing the LLM with high-level context. An appended transcript sentence or chunk for classification. This gave the model situational awareness, which helped distinguish between relevant and off-topic content more effectively.

3. Transcript Collection and Manual Labeling(human annotation): We used Whisper AI to extract transcripts from selected YouTube videos. Each sentence in these transcripts was manually labeled as: 0 = Non-Casual (relevant to the main topic) 1 = Casual (off-topic, filler, or unrelated discussion) This custom labeled dataset served as the ground truth for evaluation.

4. Casual Segment Filtering: To maintain the flow of the conversation, the minimum threshold value for a segment K was set to 3. Any sequence of 3 or more consecutive casual sentences was marked as discarded. This pre-

vented isolated casual sentences (like natural transitions) from being deleted, preserving coherence.

C. Evaluation 1. Evaluation Metric: Jaccard Similarity: For evaluating the effectiveness of our approach, we used Jaccard similarity as the primary metric. This metric quantitatively measures the degree of overlap between two sets, in our case, the set of non-casual sentences identified by human annotators and the set predicted by the LLM after processing. Each transcript was first manually labeled at the sentence level, with annotators assigning a label of 0 for non-casual (relevant) content and 1 for casual (off-topic or filler) content. The LLM, using our designed prompt, which included the video summary for contextual grounding, was tasked with performing the same classification. After classification, we applied a rule-based filter to remove segments where three or more consecutive sentences were labeled casual. We then compared the resulting filtered transcript with the manually curated version using the Jaccard formula:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

where A is the set of non-casual sentences from the human-labeled transcript, and B is the corresponding set from the LLM-processed transcript. This comparison helped us quantify how closely the model’s output aligned with human judgment, with higher scores indicating better agreement.

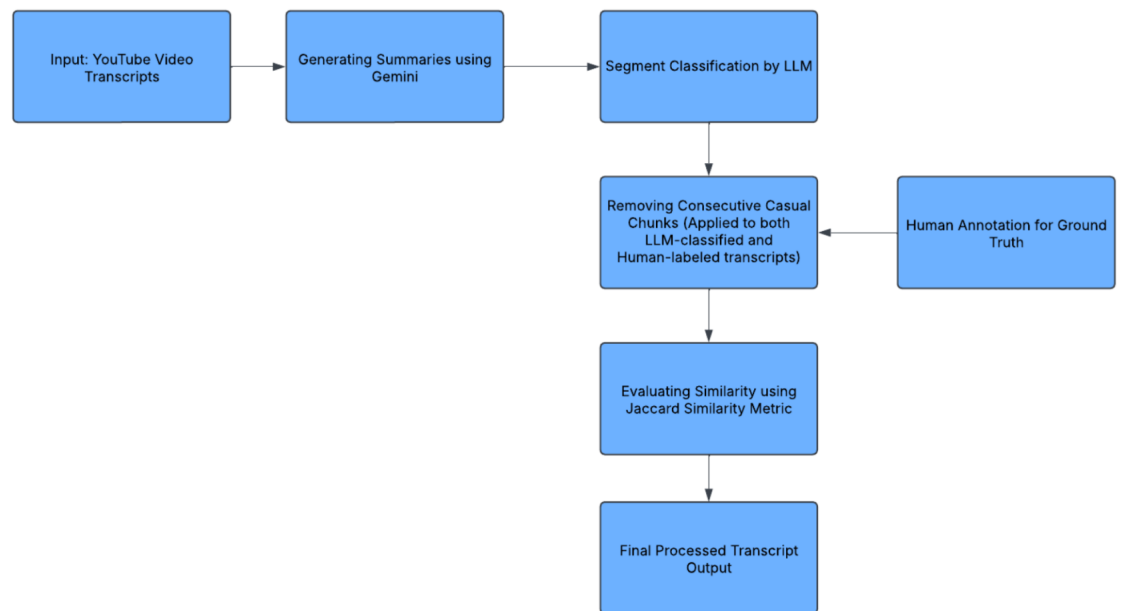


Figure 4.1: Flow Chart for Eliminating Casual Segments in Audio Transcripts

Chapter 5

Results

To evaluate our approach, we performed tests in three stages:

A. Initial CARP Model: The original CARP framework was implemented using GPT-3.5-Turbo-Instruct as a replacement for the deprecated text-davinci-003 model. It was tested on the SST-2 sentiment classification dataset, achieving an accuracy of 72.76%. The evaluation process took approximately 42 minutes. However, the performance was lower than the accuracy reported in the original paper of 93.01%, probably due to differences in model capabilities and prompt design.

B. Modified CARP Approach: To improve performance, a three-stage prompting strategy was introduced, consisting of a Clue Prompt, Reasoning Prompt, and Sentiment Prompt. This aligned more closely with CARP’s progressive reasoning technique. The modified version achieved an improved accuracy of 89.34% in the SST-2 test data set. However, this came at the cost of increased inference time (slightly longer than 1 hour) due to multiple prompt invocations.

C. Application to Audio Transcripts: For the core task of filtering casual segments from YouTube audio transcripts, we adopted a single-prompt strategy, enhanced with a Gemini-generated video summary to provide contextual grounding. Transcripts were generated using Whisper AI and manually labeled as casual or non-casual. A rule-based filter removed stretches of three or more consecutive casual sentences. The effectiveness was evaluated by computing the Jaccard Similarity between human-labeled and LLM-processed transcripts for 18 samples. The scores ranged from 0.17 to 0.72, reflecting moderate to strong agreement. This analysis demonstrates the model’s ability to eliminate irrelevant content while preserving core information, validating its use in real-world transcript refinement.

DATA	JACCARD-SCORE
TRANSCRIPT 1	0.4525
TRANSCRIPT 2	0.5324
TRANSCRIPT 3	0.4584
TRANSCRIPT 4	0.3936
TRANSCRIPT 5	0.5598
TRANSCRIPT 6	0.5926
TRANSCRIPT 7	0.2249
TRANSCRIPT 8	0.5932
TRANSCRIPT 9	0.2557
TRANSCRIPT 10	0.1758
TRANSCRIPT 11	0.1979
TRANSCRIPT 12	0.3641
TRANSCRIPT 13	0.6941
TRANSCRIPT 14	0.3562
TRANSCRIPT 15	0.3439
TRANSCRIPT 16	0.7222
TRANSCRIPT 17	0.7055
TRANSCRIPT 18	0.487

TABLE I

Figure 5.1: JACCARD SIMILARITY SCORES FOR LLM-PROCESSED AND HUMAN-ANNOTATED TRANSCRIPTS

Chapter 6

Conclusion and Future work

This project presents an automated approach for refining audio transcripts by eliminating casual conversations, thus enhancing the clarity and relevance of spoken content. Building on the CARP framework, we designed a simplified, single-prompt strategy powered by GPT-3.5-Turbo-Instruct, and further enriched it with Gemini-generated video summaries to provide necessary context during classification. The transcripts were manually labeled and filtered using a rule-based approach to remove three or more consecutive casual sentences.

To evaluate performance, Jaccard Similarity was calculated between human-labeled and model-processed outputs across 18 transcripts, showing promising alignment and demonstrating the model’s potential for structured and context-aware transcript filtering. For future work, we plan to:

- Expand the dataset with more manually annotated transcripts.
- Optimize filtering rules, especially the threshold for identifying casual segments.
- Improve model performance through prompt refinement or potential fine-tuning.

These enhancements aim to improve generalization and robustness across diverse audio content types.

References

- [1] Y. Zhang, M. Wang, C. Ren, Q. Li, P. Tiwari, B. Wang, and J. Qin, “Pushing the limit of llm capacity for text classification,” *Journal of Artificial Intelligence Research*, vol. 75, 2024.
- [2] Y. Zhang, R. Yang, X. Xu, R. Li, J. Xiao, J. Shen, and J. Han, “Teleclass: Taxonomy enrichment and llm-enhanced hierarchical text classification with minimal supervision,” *International Journal of Computer Applications*, vol. 76, 2024.
- [3] X. Sun, X. Li, J. Li, F. Wu, S. Guo, T. Zhang, and G. Wang, “Text classification via large language models,” *Findings of the Association for Computational Linguistics: EMNLP*, vol. 71, 2023.